

GAURAV GANDHI

Senior Data Scientist — Applied AI | GenAI & LLM Systems · RAG & Multi-Agent · Fine-Tuning · Evaluation | 5 Years

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PROFESSIONAL SUMMARY

Senior Data Scientist with 5 years building and shipping enterprise GenAI and machine-learning systems end-to-end — from business problem through production deployment, evaluation, and iteration. Leads and mentors a 5-person data-science team inside Uber's AI org (via Indium Software), owning the modeling roadmap, evaluation strategy, and technical review across document-intelligence, LLM fine-tuning, and multi-agent RAG workstreams. Modeling depth across transformer and diffusion architectures, two-tower/contrastive retrieval, parameter-efficient fine-tuning (LoRA/QLoRA), model calibration, and distributed training (PyTorch, Ray Train, DeepSpeed ZeRO-3), with the production engineering (Python, FastAPI, Docker, GCP) to ship models into live systems. Maintains a parallel body of independent applied-AI work, each project released with a pre-registered evaluation harness.

PROFESSIONAL EXPERIENCE

Indium Software (Client: Uber Technologies — Uber AI)

Jun 2024 – Present

Senior Data Scientist

May 2025 – Present · Bengaluru, India

- **Team leadership:** Lead and mentor a 5-person data-science team across GenAI document-intelligence and conversational-AI workstreams — owning the modeling roadmap, evaluation strategy, and technical review.
- **LLM fine-tuning for document intelligence (LoRA/QLoRA):** Raised document-extraction field-level accuracy to 97%+ by building a parameter-efficient fine-tuning framework (Llama 3.1, GPT-Neo) with a layer-freezing strategy, instruction-tuning data design, and an automated field-accuracy eval harness; mentored multiple teams on prompt and evaluator design.
- **Vision document-quality gate for auto-approval (ViT):** Removed the manual-review bottleneck in earner document verification by training a Vision Transformer that scores each upload against 9 rejection reasons (blur, glare/shadow, missing corners, damage, occlusion, and more) across 55 document types, auto-approving only below a calibrated per-(type, reason) human-recall threshold.
- **Multi-agent GenAI copilot for enterprise data (Hybrid RAG + NL2SQL):** Cut analytics turnaround from hours to seconds for non-technical teams by designing a production multi-agent copilot combining hybrid RAG (dense + sparse retrieval), NL2SQL, a schema knowledge graph, and LangChain tool orchestration — live across business units.

Data Scientist

Jun 2024 – May 2025 · Bengaluru, India

- **Document-understanding Transformer at production scale:** Owned the multi-task training objective and extraction head for an encoder-decoder Transformer pretrained on 50M+ documents across 144 A100 GPUs (PyTorch, Ray Train, DeepSpeed ZeRO-3) — shipped to production at 95%+ field accuracy, displacing manual review for \$10M+ in annual cost savings.
- **Session-aware recommendation engine (BERT + sequence modelling):** Drove a sustained 27% lift in user engagement by replacing static collaborative filtering with a BERT-based session-aware sequence model capturing evolving user intent, validated via A/B tests across releases.

FedEx Express (Consumer Logistics & Supply Chain)

Aug 2022 – Jun 2024

Decision Scientist

Bengaluru, India (Remote)

- **Probabilistic anomaly detection for account retention:** Recovered lost enterprise shipping volume — surfaced via Average-Daily-Volume decline — by deploying an ensemble of Bayesian change-point detection, probabilistic clustering, and profitability scoring, shaping regional supply strategy cross-functionally.
- **Demand forecasting & capacity optimisation (SARIMA + Bayesian tuning):** Reduced operational inefficiencies ~20% by upgrading ARIMA pipelines to SARIMA with Box-Jenkins tuning, Fourier seasonality, and automated anomaly filtering, feeding sales and staffing plans.

Tata Consultancy Services (TCS)

Jul 2021 – Jul 2022

Data Engineer

Bengaluru, India (Remote)

- **High-volume data pipelines & analytics (GCP):** Ensured reliable, scalable downstream analytics for a high-volume booking/transaction client by designing end-to-end GCP ETL pipelines plus dashboards for campaign targeting, segmentation, and trend monitoring.

APPLIED AI PROJECTS

TriagleIQ — Four-Stage ML Issue-Triage Service (GCP Cloud Run) [Live](#) · [Code](#)

- Reached **87.1% (Kubernetes) / 89.8% (VS Code) top-3 component-classification accuracy** with a temperature-calibrated classifier, BGE + FAISS similar-issue retrieval, and a LightGBM quantile estimator delivering CQR conformal intervals at 76.2% / 74.6% empirical coverage against an 80% target; LLM fabrication rate enforced as a hard CI gate with ratchet semantics.

Style Maitri — Agentic Stylist for Indian Occasion Wear [Live](#) · [Code](#)

- Hit **93.8% intent-parsing accuracy (n=211, gate 88%)** and **P@5 0.841 / NDCG@10 0.918** over a 52,494-item catalogue from 8 stores, by building a LangGraph agent on hybrid retrieval (dense FAISS + BM25 fused via Reciprocal Rank Fusion, cross-encoder rerank) with a sentence-level grounding layer that scrubs invented price/size/fabric claims; all gates enforced in CI.

AgentGauge — Causal A/B Harness for Agent Tool-Use (pip install agentgauge-harness) [PyPI](#) · [Code](#)

- Measured a **-13.3 to -28.9 pp** causal effect of a single blocking tool-description defect on agent task success across 3 model families, by building a pre-registered A/B harness with 6 model adapters, 100% replay determinism, and per-run cost accounting.
- Falsified an earlier hypothesis that an 8-axis description-quality score predicts agent success (45 tool sets), encoded 10 measurement artifacts as standing audit checks, and rebuilt the tool as a regression harness — the negative result was published rather than dropped.

Multimodal Fashion Recommender — Two-Tower Contrastive Retrieval [Demo](#) · [Code](#)

- Improved **Recall@10** by **3.06× over a popularity baseline (0.0328 vs 0.0107)** and 2.12× over co-purchase, by training a two-tower retrieval model that fuses CLIP ViT-B/32 image embeddings with SBERT text embeddings under in-batch InfoNCE contrastive loss with FAISS ANN serving, validated by a per-modality ablation.

AetherArt — Diffusion Fine-Tuning & Style-LoRA Evaluation (3 models on Hugging Face) [Live](#) · [Code](#)

- Trained and published three style LoRAs (Ukiyo-e SDXL, Ukiyo-e SD 2.1, Pattachitra SDXL) within a **6.2 GB peak-VRAM budget** on consumer hardware, composed with Hyper-SD few-step distillation and ControlNet (Canny + Depth) conditioning, served on GCP Cloud Run (L4 GPU) under a hard \$25 spend cap.
- Established that CLIP score responds to semantic sweeps but is blind to rendering-level parameters (quantization, scheduler, trigger token), documented five measurement-defect classes in style-LoRA evaluation, and reversed an earlier “LoRA loses to base” verdict via a corrected weight sweep — setting the published default to weight 0.5.

Hinglish Sentence-Embedding Model & Warmer (daily semantic word game) [Model](#) · [Live](#)

- Lifted Hinglish relatedness from **-0.003 to 0.813 Spearman correlation** by fine-tuning a sentence-transformer with LoRA (r=8) and hard-negative mining, after a nine-model bake-off and two documented failed attempts; published to Hugging Face and serving a live daily word game.

DealHunter — Multi-Agent Flight Search & Itinerary Optimisation [Live](#) · [Code](#)

- Turned free-text trip requests into two distinct Pareto-optimal itineraries (best-value vs. best-experience) by building a deterministic Planner→Optimizer multi-agent system with a ConversationManager for multi-turn refinement, an LLM-as-judge coherence harness, and SSE streaming to a Next.js UI; hardened by a **deterministic replay suite** with per-run cost accounting.

Samidha Reviews — Multilingual Review Intelligence (samidhareviews.xyz) [Code](#)

- Reached **83.8% field-level extraction accuracy across English, Hindi, and Hinglish against an 83% CI gate**, by building a tiered Groq-LLM extraction service (Llama 3.3 70B, JSON mode) with an authenticity score, prompt-injection defense, PII redaction, and multi-tenant isolation.

More on GitHub: ShelfSense (M5 demand forecasting, WRMSSE 0.5693 vs 0.8956 naive) · Gold Rate Tracker (204-fold backtest; ships the naive baseline after it beat Chronos-Bolt-Tiny on MAE, 251.99 vs 293.10, Wilcoxon p=0.0001) · tracegauge (PyPI) · Reclaim · Expense Tracker.

RESEARCH (ONGOING)

- Tool-Description Quality Is Not One Axis: A Regime Analysis of Where It Helps and Where It Backfires* — preprint, under submission. Shows tool-description quality is not a single better/worse axis: the precision that helps an agent disambiguate confusable tools is orthogonal to, or actively harmful for, context-rich selection and tool retrieval.
- Powering Agent Evaluations: Variance Structure, Measurement Artifacts, and Minimum Detectable Effects in Tool-Use Benchmarks* — preprint, under submission. Contributes variance decomposition, minimum-detectable-effect floors, and an artifact-screening protocol for agent benchmarks.
- Measurement Defects in Style-LoRA Evaluation* — in preparation, from the AetherArt evaluation programme. Continuing methodology work on agent and generative-model evaluation across AgentGauge, tracegauge, and AetherArt.

TECHNICAL SKILLS

- Machine Learning & Deep Learning:** Transformers (BERT, RoBERTa, GPT, Llama), CLIP, ViT, Diffusion (SDXL, ControlNet), CNNs, LSTMs, Sequence Modelling, Two-Tower / Contrastive Learning (InfoNCE), Recommender & Ranking Systems, Feature Engineering, Model Calibration, Conformal Prediction, A/B Testing & Experimentation, Power Analysis, Causal Inference, Bayesian Inference, Time-Series Forecasting
- LLMs, GenAI & Fine-Tuning:** LLM APIs (OpenAI, Anthropic, Gemini, Hugging Face, Groq), LoRA / QLoRA, PEFT, Instruction Tuning, Hard-Negative Mining, Prompt Engineering & LLM-as-Judge Evaluation, RLHF / DPO, Guardrails & Grounding, Structured Output / JSON Mode, Prompt-Injection Defense
- RAG, Retrieval & Agents:** RAG, Hybrid Retrieval (dense + BM25 / RRF), Cross-Encoder Reranking, Vector Search (FAISS, pgvector, Pinecone), NL2SQL, Knowledge Graphs, Multi-Agent Systems, LangChain, LangGraph, MCP
- Languages & Frameworks:** Python (Advanced), SQL, Spark SQL, PySpark, PyTorch, Hugging Face Transformers, diffusers, PEFT, sentence-transformers, scikit-learn, LightGBM, NumPy, Pandas, Polars, FastAPI, Git, Pytest, OOP
- Deployment & MLOps:** Docker, GCP (Cloud Run, Vertex AI, Workload Identity Federation), Azure, MLflow, GitHub Actions CI/CD, Dagster, Ray Train, DeepSpeed ZeRO-3, model evaluation & production monitoring

EDUCATION

Post Graduate Diploma in Data Science — Great Lakes Institute of Management

Feb 2022 – Nov 2022

GPA 8.45 / 10.0 · Chennai, India (Remote)

B.Tech, Electronics & Communication Engineering — Manipal University Jaipur

Jul 2017 – May 2021

GPA 8.05 / 10.0 · Jaipur, India

CERTIFICATIONS & CONTINUING EDUCATION

- Google Cloud Professional Data Engineer** — Google Cloud
- Deep Learning Specialization · Natural Language Processing Specialization · Generative AI with LLMs / LLMOps** — DeepLearning.AI (Coursera)
- Machine Learning with Python** — IBM (Coursera)
- Neural Networks: Zero to Hero** — Andrej Karpathy; self-paced, implementing backpropagation, language models, and a GPT from scratch (micrograd, makemore, nanoGPT).